

EIS Measurement PCB

Bill of Materials — Component Order List

Generated: 02 July 2026 | MCU: STM32F303CCT6 | Supplier: Reichelt Elektronik (DE)

Component / Part Number	Qty	Reichelt Order Link (click to open search)
Microcontrollers & Digital ICs		
STM32F303CCT6 ARM Cortex-M4F, 72 MHz, 256 KB Flash, LQFP-48 Central MCU — ADC, SPI, USART, DMA	1	https://www.reichelt.de/search.html?SEARCH=STM32F303CCT6
CP2102-GMR Silicon Labs USB-to-UART Bridge, QFN-28 Converts USB (PC) to USART1 (STM32 PA9/PA10)	1	https://www.reichelt.de/search.html?SEARCH=CP2102 ■ Also available at Mouser / DigiKey if not in stock
W25Q32JVSSIQ Winbond 32 Mbit SPI Flash, SOIC-8 Stores sweep measurement records; shares SPI1 (CS=PB12)	1	https://www.reichelt.de/search.html?SEARCH=W25Q32
DDS Signal Generator		
AD9833BRMZ Analog Devices Programmable Waveform Generator, MSOP-10 Generates swept sine; SPI-controlled (FSYNC=PA4); 25 MHz MCLK	1	https://www.reichelt.de/search.html?SEARCH=AD9833 ■ Specialty ADI part — also at Mouser / DigiKey
Analog ICs		
OPA2140AID Texas Instruments Dual Precision FET-Input Op-Amp, SOIC-8 Amp1A: excitation unity-gain buffer; Amp2A: TIA (I-to-V)	1	https://www.reichelt.de/search.html?SEARCH=OPA2140 ■ Precision FET op-amp — also at Mouser / DigiKey
ADR4533BRZ Analog Devices 3.3 V Precision Voltage Reference, SOIC-8 Feeds STM32 VDDA and VREF+ for accurate ADC full-scale	1	https://www.reichelt.de/search.html?SEARCH=ADR4533 ■ Low-drift (<25 uV/degC) reference — also at Mouser / DigiKey
MCP1700T-3302E/TT Microchip 3.3 V Ultra-Low IQ LDO Regulator, SOT-23 USB-C VBUS (5 V) to 3.3 V system supply; 250 mA output	1	https://www.reichelt.de/search.html?SEARCH=MCP1700
Oscillators & Crystals		

Component / Part Number	Qty	Reichelt Order Link (click to open search)
ASE-25.000MHZ-LC-T Abracon 25 MHz SMD Crystal Oscillator (full-can, 4-pad) AD9833 MCLK — determines DDS freq word accuracy	1	https://www.reichelt.de/search.html?SEARCH=Oszillator+25+MHz+SMD ■ 5x3.2 mm or 7x5 mm footprint — verify package on datasheet
ABM3B-8.000MHZ-B2-T Abracon 8 MHz SMD Crystal, 3.2x2.5 mm STM32 HSE (PF0/OSC_IN, PF1/OSC_OUT) -> PLL x9 = 72 MHz	1	https://www.reichelt.de/search.html?SEARCH=Quarz+8+MHz+SMD
Capacitors		
100 nF / 50 V / X7R / 0402 MLCC Decoupling Capacitor One per IC power pin: STM32 x5, AD9833 x2, OPA2140 x2, ADR4533 x1, W25Q32 x1, CP2102 x2 (order extras)	15	https://www.reichelt.de/search.html?SEARCH=100nF+0402+X7R
1 uF / 16 V / X5R / 0402 MLCC Bulk Bypass Capacitor ADR4533 output x1, MCP1700 input x1, MCP1700 output x1, STM32 VDDA x1, extra x1	5	https://www.reichelt.de/search.html?SEARCH=1uF+0402+X5R
10 uF / 16 V / X5R / 0805 MLCC Bulk Input Capacitor USB-C VBUS input bulk — absorbs cable inductance transients	1	https://www.reichelt.de/search.html?SEARCH=10uF+0805+X5R
22 pF / 50 V / C0G / 0402 MLCC Crystal Load Capacitor Two load caps for 8 MHz STM32 crystal (C_L = 10 pF typ.)	2	https://www.reichelt.de/search.html?SEARCH=22pF+0402+C0G ■ Verify exact C_L value from ABM3B-8.000 datasheet
Resistors (all 0402, 1%)		
330 Ohm / 0402 LED Current-Limiting Resistor Series resistor for status LED on PB13 (3.3 V - V_f) / I_f	1	https://www.reichelt.de/search.html?SEARCH=330+Ohm+0402
5.1 kOhm / 0402 USB-C CC Pull-Down Resistors CC1 and CC2 pins — configures board as 5 V sink (UFP)	2	https://www.reichelt.de/search.html?SEARCH=5%2C1+kOhm+0402
10 kOhm / 0402 General Pull-Up / Pull-Down Resistors W25Q32 /CS pull-up (PB12), BOOT0 pull-down, TIA Rf (default)	3	https://www.reichelt.de/search.html?SEARCH=10+kOhm+0402 ■ Rf is user-selectable — add 1k / 100k variants as needed
LED		

Component / Part Number	Qty	Reichelt Order Link (click to open search)
Green LED / 0402 SMD Status / Activity LED Connected to PB13 (active high); toggled once per sweep step	1	https://www.reichelt.de/search.html?SEARCH=LED+gruen+0402+SMD
Connectors		
USB-C Receptacle (SMD) USB Type-C Female Connector — Power + Data VBUS (5 V in), D+/D- to CP2102, CC1/CC2 pull-downs	1	https://www.reichelt.de/search.html?SEARCH=USB-C+Buchse+SMD ■ <i>Verify footprint matches your PCB layout</i>
Electrode Connector (2x BNC PCB) Coaxial BNC PCB-Mount Connector Electrode+ (excitation) and Electrode- (return / TIA input)	2	https://www.reichelt.de/search.html?SEARCH=BNC+Buchse+Leiterplatte ■ <i>Alternatively use 2-pin screw terminal or SMA connector</i>
SWD Debug Header — 2x5 / 1.27 mm ARM Cortex Debug Connector (Cortex-M 10-pin) SWDIO (PA13), SWDCLK (PA14), SWO, RESET, VDD, GND	1	https://www.reichelt.de/search.html?SEARCH=Stiftleiste+2x5+1%2C27
PCB		
Custom PCB (2-layer or 4-layer) Bare PCB manufactured from your KiCad / Altium design files Min. 4/4 mil clearance recommended; controlled impedance for SPI	1	https://jlcpcb.com ■ <i>Order from JLCPCB, PCBWay, or Eurocircuits (EU lead time ~5d)</i>

Notes

1. All Reichelt links open a search results page — verify part number and package before adding to cart.
2. Specialty precision ICs (AD9833, OPA2140, ADR4533) may not be stocked by Reichelt. Order from [Mouser Deutschland](#) or [DigiKey Germany](#) if unavailable.
3. Resistor R_f (TIA feedback) default is 10 kOhm. For higher-impedance cells order 100 kOhm (0402) as well.
4. Crystal load capacitors (22 pF): verify the exact C_L value from the ABM3B-8.000MHZ datasheet (specified C_L is 10 pF; board caps = 2*(C_{stray} + C_{board}) to match C_L).
5. PCB: JLCPCB (<https://jlcpcb.com>) ships to Germany in 7-10 days at low cost. Eurocircuits (<https://www.eurocircuits.com>) offers EU-manufactured boards in ~5 days.
6. All SMD passives are 0402 footprint. A solder paste stencil is strongly recommended.
7. The electrode connector type (BNC, SMA, or screw terminal) depends on your measurement setup.